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Supporting Information

The Permeation Mechanism of Cisplatin Through a Dioleoylphosphocholine Bilayer**

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S1 Umbrella Sampling Set Up

It is well known that the number of windows and the force constant of the bias harmonic potential must be properly chosen to have a good overlap among the reaction-coordinate probability distributions of consecutive windows. Such a good overlap is required to obtain an accurate free-energy profile along the reaction coordinate, also called potential of mean force, from the biased probability distributions, especially when the Weighted Histogram Analysis Method (WHAM) approach is employed.[1] Figure S1 shows the computed probability distributions for the umbrella sampling windows centred at 10.0, 10.5, 11.0, 11.5 and 12.0 Å as example windows. It can be seen that the distributions of neighbouring windows present a strong overlap, which indicates a good sampling along the reaction coordinate and validates the choice of the simulation parameters employed.

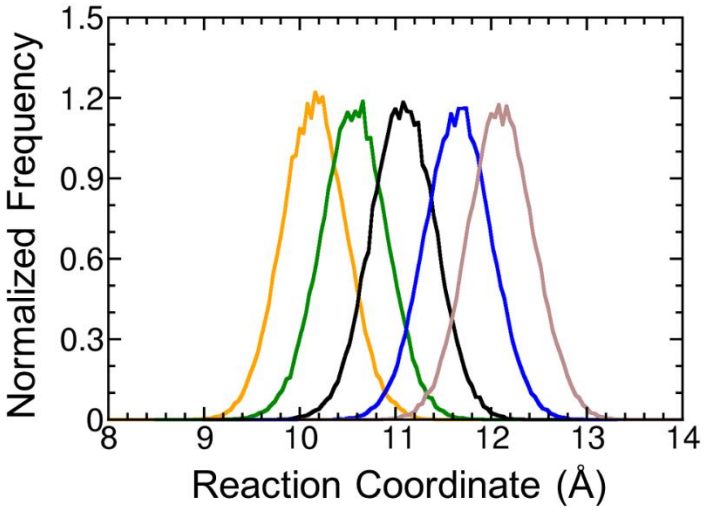


Figure S1. Reaction coordinate probability distributions for the windows centred at 10.0, 10.5, 11.0, 11.5 and 12.0 Å.

S2 Permeability Coefficient

The permeability coefficient of 3.6×10^{-7} ms^{-1} was computed upon integration of the resistance coefficient along the permeation pathway. In turn, the resistance coefficient can be computed from the Z-dependent diffusion coefficient (plotted in Figure S2) and from the potential of mean force (plotted in Figure 2B). The details about the full protocol are discussed elsewhere.[2]

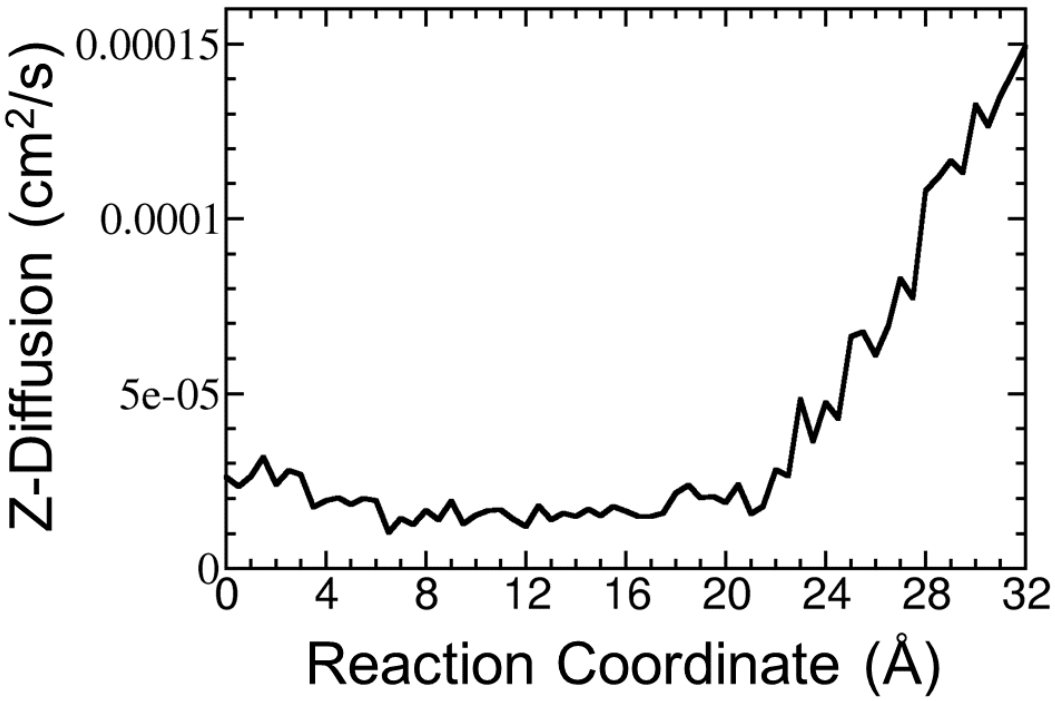


Figure S2. Z-dependent diffusion coefficient of cisplatin.

S3 Force Field Parameters

Table S1. Force field parameters for cisplatin: bond force constants k_b, equilibrium distances R_0, angle force constants k_a, equilibrium angles θ_0, atomic charges q and Lennard-Jones parameters $R_{min}/2$ and ε.

Bond Parameters			
Bond	k_b (kcal/[mol Å^2])	R_0 (Å)	
Pt-Cl	123.100	2.305	
Pt-N	88.900	2.111	
N-H	392.400	1.019	
Angle Parameters			
Angle	k_a (kcal/[mol rad^2])	θ_0 (degrees)	
N-Pt-H	132.790	98.410	
Cl-Pt-N_{cis}	105.950	83.010	
Cl-Pt-N_{trans}	106.350	178.580	
Cl-Pt-Cl	88.200	95.570	
Pt-N-H	52.880	110.140	
H-N-H	41.400	106.40	
Intermolecular parameters			
Atom	q (a.u.)	$R_{min}/2$ (Å)	ε (kcal/mol)
Pt	0.069	2.053	1.055
Cl	-0.375	2.597	0.038
N	-0.410	1.896	0.045
H1[a]	0.239	0.052	0.018
H2[b]	0.256	0.052	0.018

[a]Hydrogen atoms that do not belong to the plane formed by N(Cl)-Pt-N(Cl).

[b]Hydrogen atoms that belong to the plane formed by N(Cl)-Pt-N(Cl).

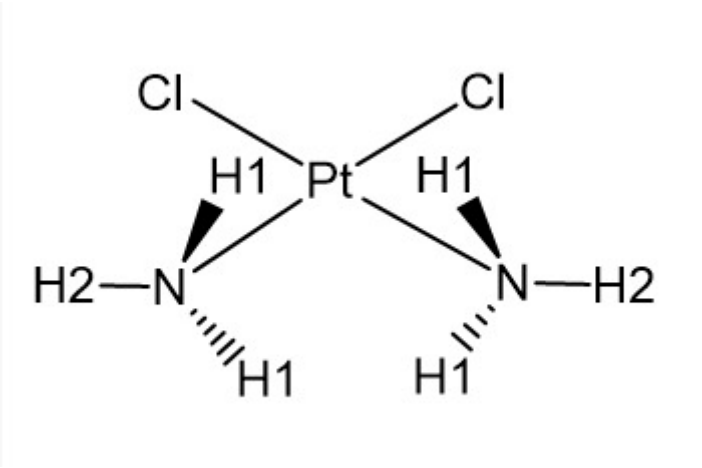


Figure S3. Atom labels of cisplatin employed in Table S1.

S4 References

[1] J. Kästner, *WIREs: Comput. Mol. Sci.* **2011**, *1*, 932-942.

[2] C. T. Lee, J. Comer, C. Herndon, N. Leung, A. Pavlova, R. V. Swift, C. Tung, C. N. Rowley, R. E. Amaro, C. Chipot, Y. Wang, J. C. Gumbart, *J. Chem. Inf. Model.* **2016**, *56*, 721-733.